1 Find the value of 3^{-1}

(1 mark)

2 Find the value of $\left(\frac{4}{5}\right)^{-1}$

(1 mark)

3 Find the value of 5^{-1}

(1 mark)

4 Find the reciprocal of 3

(1 mark)

5 Find the value of $100^{\frac{1}{2}}$

(1 mark)

6 Find the value of $64^{\frac{1}{2}}$

(1 mark)

7 Find the value of $49^{\frac{1}{2}}$

(1 mark)

8 Find the value of $81^{\frac{1}{2}}$

(1 mark)

9 Find the value of $36^{-\frac{1}{2}}$

(1 mark)

10 Find the value of $64^{\frac{1}{3}}$

(1 mark)

11 Find the value of $8^{\frac{1}{3}}$

(1 mark)

12 Find the value of $27^{\frac{1}{3}}$

(1 mark)

13 Find the value of $125^{\frac{1}{3}}$

(1 mark)

14 Find the value of $64^{-\frac{1}{3}}$

(1 mark)

15 Find the value of $64^{-\frac{2}{3}}$

(2 mark)

16 Find the value of $125^{\frac{2}{3}}$

(2 mark)

17 Find the value of $8^{-\frac{2}{3}}$

(2 mark)

18 Find the value of $27^{-\frac{2}{3}}$

(2 mark)

19 Find the value of $(8x^6)^{\frac{2}{3}}$

(2 mark)

20 Find the value of $\left(\frac{64}{125}\right)^{-\frac{2}{3}}$

(2 mark)

21 Find the value of $\left(\frac{25}{16}\right)^{-\frac{3}{2}}$

(2 mark)

22 Find the value of $\left(\frac{8}{27}\right)^{-\frac{2}{3}}$

(2 mark)

23 Find the value of $\left(\frac{9}{4}\right)^{-\frac{3}{2}}$

(2 mark)

24 Find the value of $\sqrt[4]{2 \times 8 \times 10^{12}}$

(2 mark)

25 Find the value of $\sqrt[3]{4 \times 16 \times 10^{15}}$

(2 mark)

26 Given that $3 \times \sqrt{3} = 3^n$
Find the value n.

(2 mark)

27 Given that $3 \times \sqrt{27} = 3^n$
Find the value n.

(2 mark)

28 Given that $x = 2^p$ and $y = 2^q$
Express in terms of x and/or y,

(i) 2^{p+q}

(ii) 2^{2p}

(iii) 2^{q-1}

(3 mark)

29 Given that $3^{-n} = 0.2$
Find the value of $(3^n)^2$

(2 mark)

30 Given that $5^{-n} = 0.5$
Find the value of $(5^n)^3$

(2 mark)

31 Given that $4^n = 8$
Find the value of n.

(2 mark)

32 Given that $4^{-n} = 32$
Find the value of n.

(2 mark)